

MONOGRAPH

Mathematical, Physical, and Computational Principles of Rotational Thromboelastometry

An Open-Source Computational Framework

Mohammed W. Hammami

Independent Researcher

August 2026

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Abstract

Rotational thromboelastometry (ROTEM) is a viscoelastic assay used to evaluate whole blood coagulation dynamics in clinical settings. Despite its utility, commercial ROTEM analyzers operate as proprietary systems, obscuring the mechanical and computational algorithms that translate clot formation into viscoelastic amplitude. This opacity limits mechanistic research and educational applications. We present a transparent, open-source computational framework that models the biochemical and mechanical principles of rotational thromboelastometry. A reduced-order ordinary differential equation (ODE) system was developed to simulate thrombin generation, fibrin polymerization, fibrinolysis, and platelet activation. Biochemical states were mapped to viscoelastic amplitude using an additive Clot Elasticity (CE) transformation, incorporating a fibrin scaffold gate to ensure biomechanical fidelity during lysis. The model was integrated numerically using a stiff solver (LSODA) with strict mass conservation tolerances. Simulated assays (EXTEM, FIBTEM, APTEM) were calibrated against clinical reference ranges and validated phenotypically. The framework accurately reproduces normal clotting times and maximum clot firmness. Perturbations of initial state variables successfully simulate hypofibrinogenemia, thrombocytopenia, and the lethal hyperfibrinolytic phenotype. The APTEM assay simulation correctly reverses hyperfibrinolytic decay via plasmin inhibition. This monograph and its computational implementation provide a mathematically rigorous, open-source laboratory for investigating viscoelastic coagulation testing.

Keywords: rotational thromboelastometry; computational hematology; ordinary differential equations; viscoelasticity; hyperfibrinolysis; reduced-order model.

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General Introduction

Viscoelastic testing by rotational thromboelastometry (ROTEM) provides a dynamic, whole-blood assessment of coagulation, fibrin polymerization, and fibrinolysis. Unlike conventional plasma assays, ROTEM captures the integrated contributions of cellular and plasma components, making it indispensable for managing trauma-induced coagulopathy, perioperative bleeding, and massive transfusion protocols. However, commercial ROTEM devices function as proprietary measurement systems. The internal mechanical transduction, signal processing algorithms, and proprietary smoothing kernels remain undisclosed, effectively treating the analyzer as a black box. This opacity prevents independent verification of how biochemical states map to reported clinical parameters and complicates mechanistic research.

Computational modeling offers a pathway to deconstruct this black box. Comprehensive mathematical models of the coagulation cascade, such as the 34-equation Hockin-Mann system, accurately describe the stoichiometric regulation of thrombin generation. However, the clinical utility of such exhaustive systems is limited by computational expense, identifiability issues, and extensive inter-patient parameter variability. Reduced-order kinetic models collapse these complex cascades into fewer governing ordinary differential equations (ODEs) while preserving the essential biophysical phases of initiation, propagation, and termination. Such models are highly suitable for real-time simulation and educational translation.

This monograph establishes the mathematical, physical, and computational principles of a reduced-order ROTEM simulation framework. We define the governing ODEs for the core hemostatic network, including thrombin generation, fibrinogen conversion, plasmin-mediated degradation, and platelet activation. We then specify a mechanical mapping that translates these biochemical states into viscoelastic amplitude via Clot Elasticity (CE) space. Assay-specific perturbations (EXTEM, FIBTEM, APTEM) and pathological presets (hypofibrinogenemia, hyperfibrinolysis) are mathematically defined as modifications to state variables and rate constants. Finally, we detail the numerical implementation, software architecture, and phenotypic validation of the simulator. By providing an open-source, mechanistic framework, this work aims to serve as a computational laboratory for hypothesis generation, assay optimization, and educational exploration of viscoelastic hemostasis.

Mathematical Specification

2.1 Introduction to the Reduced-Order Modeling Approach

The mathematical representation of blood coagulation as a system of coupled ordinary differential equations (ODEs) has been pursued through both comprehensive mechanistic frameworks and reduced-order approximations. Comprehensive models, such as the 34-equation, 42-rate-constant system developed by Hockin et al. [4], describe 27 independent equilibrium expressions governing the fates of 34 chemical species. While such systems capture the stoichiometric regulation of the extrinsic pathway in considerable detail, their clinical utility is constrained by the large number of unknown or poorly constrained rate constants, the computational expense of numerical integration, and the substantial inter-patient variability in both protein concentrations and kinetic parameters [1, 8].

The demand for patient-specific computational models with readily obtainable parameters has motivated the development of reduced-order kinetic representations. As articulated by Bannoud et al. [1], there exists a critical gap in the availability of computational models based on phenomenological or first-principles approaches that effectively incorporate affordable and routinely used clinical test results for predicting blood coagulation. Their systematic review of 53 studies revealed that kinetic parameters governing clot formation dynamics exhibit variability of up to 1,000-fold across investigations, complicating the calibration of large-scale models and hindering translation to clinical practice.

Ratto et al. [8] proposed a methodological approach to patient-specific modeling that begins with conventional thrombin generation tests to determine the parameters of a reduced kinetic model comprising three ODEs with nine kinetic constants. This simplification was justified by the observation that complete kinetic systems contain too many equations and unknown constants, rendering the inverse problem for parameter determination highly under-determined when experimental data are limited to thrombin generation curves. The reduced model preserves the essential biophysical structure of the coagulation cascade, initiation, propagation, and termination phases, while permitting parameter identification from clinically accessible assays [8].

The rationale for model reduction is supported by the sensitivity analysis of Luan et al. [5], who demonstrated that a 92-protein, 148-interaction model of the coagulation cascade could be computationally screened for points of fragility using Monte Carlo sensitivity analysis. Their calculation of overall state sensitivity coefficients (OSSCs) revealed that only a subset of mechanisms, primarily those involving fluid and surface phase factor X/activated factor X

activity and thrombin-mediated platelet activation, were fragile, while mechanisms involving factor IX/activated factor IX and factor VIII/activated factor VIII were robust [5]. This finding implies that many parameters in comprehensive models contribute negligibly to macroscopic predictions and can be eliminated or aggregated without substantial loss of predictive capacity, thereby justifying the collapse of complex cascades into fewer governing equations.

2.2 Biochemical Reaction Network

2.2.1 Thrombin Generation

Tissue factor (TF) initiates the generation of thrombin (IIa) through the assembly of the extrinsic tenase complex (TF·VIIa), which activates factors IX and X, ultimately leading to the formation of the prothrombinase complex (Xa·Va) that converts prothrombin (II) to thrombin [4]. The Hockin et al. [4] model describes this process through 34 differential equations with 42 rate constants, representing 27 independent equilibrium expressions. The generation of thrombin proceeds through operationally defined initiation, propagation, and termination phases, displaying a nonlinear dependence on TF, antithrombin III (AT-III), and tissue factor pathway inhibitor (TFPI) concentrations, with the combination of these inhibitors exhibiting kinetic thresholds [4].

For the purposes of reduced-order specification, thrombin generation can be formulated using Michaelis-Menten kinetics for the prothrombinase-catalyzed conversion of prothrombin, supplemented by a direct factor Xa-membrane activation pathway and an inhibition term representing AT-III-mediated inactivation. The prothrombinase complex catalyzes the conversion of prothrombin to meizothrombin with a catalytic rate constant $k_{cat}^{prot} = 63.5 \text{ s}^{-1}$ [4], followed by rapid conversion of meizothrombin to thrombin. Direct activation of prothrombin by factor Xa-membrane in the absence of factor Va proceeds with $k_{cat}^{Xa} = 2.3 \times 10^{-3} \text{ s}^{-1}$ and $K_m^{Xa} = 0.3 \text{ } \mu\text{M}$ [4]. The second-order rate constant for thrombin inactivation by AT-III is $k_{AT} = 7.1 \times 10^3 \text{ M}^{-1}\text{s}^{-1}$ [4].

The governing ODE for thrombin concentration is:

$$\frac{d[\text{IIa}]}{dt} = k_{cat}^{prot} \cdot [\text{Xa} \cdot \text{Va}] \cdot \frac{[\text{II}]}{[\text{II}] + K_m^{prot}} + k_{cat}^{Xa} \cdot [\text{Xa}] \cdot \frac{[\text{II}]}{[\text{II}] + K_m^{Xa}} - k_{AT} \cdot [\text{IIa}] \cdot [\text{AT-III}]$$

where $[\text{Xa} \cdot \text{Va}]$ is the prothrombinase complex concentration, $[\text{Xa}]$ is the free factor Xa concentration, $[\text{II}]$ is the prothrombin concentration, K_m^{prot} is the Michaelis constant for prothrombinase, and $[\text{AT-III}]$ is the antithrombin III concentration. The first term represents prothrombinase-mediated activation, the second term represents direct Xa-membrane activation, and the third term represents irreversible inhibition by AT-III. This formulation captures the biphasic behavior of thrombin generation, an initiation phase during which factor Xa-membrane activity predominates, followed by a propagation phase dominated by prothrombinase, as described in the Hockin et al. [4] model.

2.2.2 Fibrinogen Conversion to Polymerized Fibrin

The conversion of fibrinogen to polymerized fibrin proceeds through a multi-step mechanism: fibrinopeptide A cleavage by thrombin, protofibril formation via half-staggered monomer association, and lateral aggregation of protofibrils to form fibers [9]. Weisel and Nagaswami [9] formulated differential equations describing the kinetics of polymerization, correlating model outputs with turbidity profiles and electron microscopic observations.

The simplest model considers three sequential processes with associated rate constants: fibrinopeptide A cleavage (k_A), protofibril initiation and growth (k_{pi} , k_{pg}), and fiber initiation and growth (k_{fi} , k_{fg}). The governing equations for fibrinogen (Fg), fibrin monomer (f), protofibrils (f_n), and fibers (f_r) are:

$$\begin{aligned}\frac{d[\text{Fg}]}{dt} &= -k_A \cdot [\text{Fg}] \\ \frac{d[f]}{dt} &= k_A \cdot [\text{Fg}] - k_{pi} \cdot [f]^2 - k_{pg} \cdot [f] \cdot [f_n] \\ \frac{d[f_n]}{dt} &= k_{pi} \cdot [f]^2 - 2 \cdot k_{fi} \cdot [f_n]^2 - k_{fg} \cdot [f_r] \cdot [f_n] \\ \frac{d[f_r]}{dt} &= 2 \cdot k_{fi} \cdot [f_n]^2\end{aligned}$$

where k_A is the rate of fibrinopeptide A cleavage, k_{pi} is the rate constant for protofibril initiation, k_{pg} is the rate constant for protofibril growth, k_{fi} is the rate constant for fiber initiation, and k_{fg} is the rate constant for fiber growth by protofibril addition [9]. The requirement that protofibrils reach a minimum length of 10–20 monomers before lateral aggregation could occur was necessary to reproduce the lag period observed experimentally in turbidity profiles [9]. The rate constant for fiber growth must exceed that for fiber initiation ($k_{fg} > k_{fi}$) to yield fiber diameters consistent with electron microscopic observations [9].

For integration with the thrombin generation ODE, the fibrinogen consumption and fibrin production can be expressed as thrombin-catalyzed reactions:

$$\begin{aligned}\frac{d[\text{Fg}]}{dt} &= -\alpha_1 \cdot [\text{IIa}] \cdot [\text{Fg}] \\ \frac{d[F]}{dt} &= \alpha_1 \cdot [\text{IIa}] \cdot [\text{Fg}] - \alpha_2 \cdot [F] \\ \frac{d[F_p]}{dt} &= \alpha_2 \cdot [F]\end{aligned}$$

where $[F]$ is the fibrin monomer concentration, $[F_p]$ is the polymerized fibrin concentration, α_1 is the rate constant for thrombin-catalyzed fibrinogen cleavage, and α_2 is the rate constant for fibrin polymerization [8]. The polymerized fibrin is assumed to be immobile and does not undergo diffusion or convection [8].

2.2.3 Plasmin-Mediated Fibrin Degradation

The degradation of fibrin by plasmin operates at the scale of individual fibrin fibers. Ouedraogo et al. [7] developed a two-dimensional stochastic model of a fibrin fiber cross-section, employing the Gillespie algorithm to study single-fiber lysis initiated by plasmin. The model tracks 11 reactions grouped into four categories: exposure of cryptic binding doublets by plasmin, unbinding of plasmin from doublets, degradation of doublets by plasmin, and crawling of plasmin between protofibrils [7].

The system state is described by the vector $\mathbf{x} = [N_{00}, N_{03}, \phi_{00}, \phi_{03}, N]$, where N_{00} is an exposed doublet with nothing bound, N_{03} is an exposed doublet with plasmin bound, ϕ_{00} is a degraded doublet with nothing bound, ϕ_{03} is a degraded doublet with plasmin bound, and N is a cryptic (unexposed) doublet [7]. The propensity functions for the degradation reactions are:

$$a_2 = k_{deg} \cdot x_1 \cdot n$$

$$a_3 = k_{deg} \cdot x_2 \cdot n$$

where k_{deg} is the plasmin-mediated rate of fibrin degradation, $x_1 = [N_{00}]$, $x_2 = [N_{03}]$, and $n = 1$ if plasmin is present at the protofibril cross-section and $n = 0$ otherwise [7]. The exposure reaction propensity is:

$$a_5 = k_{exp} \cdot x_5 \cdot n$$

where k_{exp} is the rate of cryptic doublet exposure and $x_5 = [N]$ [7]. Baseline parameter values are $k_{unbind} = 0.05 \text{ s}^{-1}$, $k_{crawl} = 57.6 \text{ s}^{-1}$, $k_{deg} = 5 \text{ s}^{-1}$, and $k_{exp} = 5 \text{ s}^{-1}$ [7].

The deterministic equivalent for the rate of intact fibrin degradation is:

$$\frac{d[F_{intact}]}{dt} = -k_{deg} \cdot [\text{Pln}] \cdot ([N_{00}] + [N_{03}])$$

where $[\text{Pln}]$ is the plasmin concentration and the term $([N_{00}] + [N_{03}])$ represents the concentration of exposed, undegraded binding sites accessible to plasmin. The fiber is considered cleaved when $\frac{2}{3}$ of the binding doublets within the cross-section have been degraded [7]. Model-experiment comparison indicated that the degradation rate should satisfy $k_{deg} \leq 10 \text{ s}^{-1}$ and that the crawling rate, if plasmin crawling occurs, should range between 10 s^{-1} and 60 s^{-1} [7].

For the purposes of real-time deterministic simulation in the computational implementation (Chapter 4), the 2-dimensional stochastic doublet states (N_{00} , N_{03}) are mathematically collapsed into a single macro-state deterministic equivalent governing the rate of intact fibrin degradation. The term representing exposed, undegraded binding sites is simplified to the total intact polymerized fibrin concentration $[F_p]$, yielding the governing ODE utilized in the software architecture:

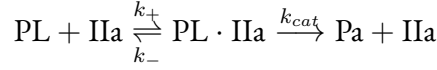
$$\frac{d[F_p]}{dt} = \alpha_2 \cdot [F] - k_{deg} \cdot [\text{Pln}] \cdot [F_p]$$

2.2.4 Platelet Activation by Thrombin

Thrombin irreversibly activates platelets through protease-activated receptors on the platelet surface [3]. In the mathematical model of Fogelson et al. [3], platelet activation by thrombin is

represented as an enzyme-catalyzed reaction following Michaelis-Menten kinetics, with distinct rate constants for fluid-phase and subendothelial-bound platelets.

The reaction scheme for platelet activation is:



where PL is the unactivated platelet, Pa is the activated platelet, k_+ is the association rate constant, k_- is the dissociation rate constant, and k_{cat} is the catalytic rate constant for activation [5]. For subendothelial-bound platelets, the corresponding parameters are $k_+^{sub} = 3.0 \times 10^{-2} \text{ nM}^{-1}\text{s}^{-1}$, $k_-^{sub} = 1.0 \times 10^{-2} \text{ s}^{-1}$, and $k_{cat}^{sub} = 9.0 \times 10^{-3} \text{ s}^{-1}$ [5].

Under the quasi-steady-state approximation, the rate of platelet activation is:

$$\frac{d[\text{Pa}]}{dt} = \frac{k_{cat} \cdot [\text{IIa}] \cdot [\text{PL}_{total}]}{K_M + [\text{IIa}]}$$

where $[\text{PL}_{total}]$ is the total concentration of activatable platelets and $K_M = (k_- + k_{cat})/k_+$ is the Michaelis-Menten constant. For subendothelial-bound platelets, $K_M^{sub} = (1.0 \times 10^{-2} + 9.0 \times 10^{-3})/(3.0 \times 10^{-2}) \approx 0.63 \text{ nM}$, indicating high-affinity activation at the injury site. The model of Fogelson et al. [3] predicts that platelet count strongly scales clot strength and that the sensitivity of thrombin production to factor XI depends on the near-wall platelet concentration, with higher platelet concentrations accelerating subendothelial coverage and thereby increasing the importance of the thrombin–factor XIa–factor IXa feedback loop.

2.3 Mechanical-to-Signal Mapping

The translation of biochemical states into the viscoelastic amplitude $A(t)$ measured in millimeters by rotational thromboelastometry requires a mapping function that captures the dependence of clot mechanical properties on both fibrin polymer concentration and platelet activation state. As demonstrated by Fogelson et al. [3], platelet count strongly scales clot strength, and as shown by Weisel and Nagaswami [9], clot turbidity, a surrogate for mechanical integrity, is proportional to the average cross-sectional area of fibrin fibers, which is itself kinetically determined by the rates of protofibril formation and lateral aggregation. The fibrin polymer concentration $[F_p]$ determines the structural framework of the clot [8], while activated platelets provide additional mechanical rigidity through platelet–fibrin interactions and platelet aggregation.

Because clot strength is linearly proportional to the sum of its structural components in the elasticity domain (as established by Solomon et al. [?] and detailed in Chapter 3.2), the state variables are first mapped to Clot Elasticity (CE) space and added, before being inverted to amplitude $A(t)$. The fibrin and platelet contributions are formulated as saturable functions in CE space:

$$CE_{fibrin}(t) = CE_{max_{fib}} \cdot \frac{[F_p](t)}{[F_p](t) + K_F}$$

$$CE_{platelet}(t) = CE_{max_{plt}} \cdot \frac{[\text{Pa}](t)}{[\text{Pa}](t) + K_{Pa}} \cdot \left(\frac{[F_p](t)}{[F_p](t) + K_{gate}} \right)$$

where $CE_{max_{fib}}$ and $CE_{max_{plt}}$ are the maximum elasticity contributions, K_F and K_{Pa} are the half-saturation constants, and K_{gate} is the fibrin scaffold gating constant. The total clot elasticity is the additive sum:

$$CE_{total}(t) = CE_{fibrin}(t) + CE_{platelet}(t)$$

The viscoelastic amplitude $A(t)$ is then derived by applying the inverse of the CE-to-Amplitude transformation:

$$A(t) = \frac{100 \cdot CE_{total}(t)}{100 + CE_{total}(t)}$$

The additive form in elasticity space reflects the mechanistic requirement that fibrin polymer and activated platelets contribute independently to the total structural integrity of the clot. Furthermore, the inclusion of the fibrin scaffold gate (K_{gate}) in the platelet term ensures that the platelet contribution to clot elasticity collapses if the underlying fibrin mesh is degraded by plasmin. Mechanistically, a platelet aggregate with no fibrin mesh holding it in the ROTEM cup has no clot integrity; thus, if $[F_p] \rightarrow 0$, $CE_{platelet} \rightarrow 0$, accurately simulating the rapid amplitude decay seen in hyperfibrinolysis. This formulation also correctly isolates the fibrin component in the FIBTEM assay, where $[Pa] = 0$, allowing $A(t)$ to be driven entirely by CE_{fibrin} .

2.4 Parameter Provenance and Uncertainty

Deterministic mathematical models of biochemical processes operate under the assumption that empirically derived rate constants governing the dynamics are known with certainty [2]. In practice, compilations of rate constants for a given reaction typically yield a range of values across laboratories, with sources of variation including protein origin, specific activity, buffer composition, and the surface composition upon which catalytic complexes assemble [2].

Danforth et al. [2] systematically evaluated the sensitivity of the Hockin et al. [4] model to perturbation of its 44 rate constants across a range from 10% to 1000% of accepted values. Using a normalized ensemble standard deviation and a time-averaged coefficient of variation over a 20-minute simulation, the analysis identified five rate constants, those involved in the regulation of the formation and function of the factor VIIa–tissue factor complex, that accounted for 50% of the aggregate variation in model predictions across all 34 species [2]. Twenty-four rate constants collectively contributed less than 13% of the total variance. When thrombin was the monitored output, the rate constant governing thrombin inhibition by antithrombin III (k_{41}) was the single most influential parameter, with uncertainty in its value producing effects equivalent to the combined contribution of the 19 least influential rate constants [2]. These findings indicate that improvements in measurement accuracy of a small subset of rate constants, particularly those governing the initiating TF·VIIa complex and thrombin inhibition, would yield the greatest increase in model predictive capability.

The sensitivity of thrombin generation to perturbations in the biochemical environment is exemplified by the effects of acidosis. Mitrophanov et al. [6] extended the Hockin-Mann model to represent pH-dependent modulation of enzymatic reaction rates by adjusting catalytic

rate constants according to $k(\text{pH}) = f(\text{pH}) \cdot k(7.4)$, where $f(\text{pH})$ is a pH-dependent scaling factor. Computational simulations demonstrated that acidosis in the range of pH 6.9 to 7.3 progressively increased clotting time, thrombin peak time, area under the thrombin trajectory, and prothrombin time, while decreasing the maximum slope of the thrombin curve and thrombin peak height ($P < 10^{-5}$) [6]. The maximum slope of the thrombin curve exhibited the highest sensitivity to acidosis at the individual-subject level, while the area under the curve displayed the smallest relative changes [6].

Analysis of 472 virtual subjects from the Leiden Thrombophilia Study revealed that, while acidosis-induced effects were consistent across the population, intersubject variability in clotting factor composition influenced the discriminative capacity of individual thrombin generation parameters. The maximum slope of the thrombin curve, despite its high intrasubject sensitivity, was inferior to clotting time, thrombin peak time, and prothrombin time as a population-level indicator of acidosis due to large intersubject variability causing distribution overlaps at distinct pH levels [6]. These results demonstrate that parameter uncertainty and environmental perturbations interact in complex ways: the parameters most sensitive to a given perturbation at the individual level may not be the most informative at the population level, and the identification of robust diagnostic indicators requires consideration of both intrasubject sensitivity and intersubject variability.

Because the reduced-order model operates in normalized concentration spaces rather than strict Molar units to maintain numerical stability within the ODE solver, applying raw literature second-order rate constants (e.g., $k_{AT} = 7.1 \times 10^3 \text{ M}^{-1}\text{s}^{-1}$) would result in instantaneous numerical exhaustion of the system. Therefore, effective rate constants are calibrated to preserve the proportional relationships between inhibition and generation, ensuring that the phenotypic behavior of the cascade is conserved without encountering floating-point underflow or excessive mathematical stiffness. This approach is consistent with the reduced-order modeling philosophy, which prioritizes macroscopic phenotypic fidelity over microscopic parametric exactness.

Assay-Specific Mathematical Definitions and Pathological Presets

3.1 Introduction to Assay Modeling

In the mathematical formulation of rotational thromboelastometry (ROTEM), the extrinsic activation and subsequent viscoelastic transduction of whole blood coagulation establish a baseline system of ordinary differential equations (ODEs). The distinct clinical assays, namely EXTEM, FIBTEM, and APTEM, are not separate biological models but rather experimental perturbations imposed upon the identical baseline state variables and kinetic rate constants defined in Chapter 2. These perturbations manifest as selective pharmacological inhibitions or pathway-specific activations that mathematically zero out targeted state variables or modify specific rate constants. This assay-specific parameterization allows the isolation of individual hemostatic components, such as fibrinogen polymerization, platelet contractile force, or plasmin-mediated proteolysis, within the unified ODE framework.

3.2 EXTEM and FIBTEM Definitions

The EXTEM assay mathematically initiates the coagulation cascade by introducing tissue factor as an exogenous activator, establishing the initial condition for the extrinsic pathway ODEs [10, 11]. Under this activation, the evolution of clot firmness reflects the integrated contributions of thrombin generation, fibrinogen polymerization, and platelet activation. To isolate the fibrinogen component, the FIBTEM assay introduces cytochalasin D, an actin polymerization inhibitor, which functionally abolishes the platelet contribution to clot strength [12]. Mathematically, this perturbation sets the platelet activation state variable, denoted as P_a , to zero.

By nullifying P_a , the resulting clot elasticity (CE) is governed entirely by the fibrinogen concentration variable. The relationship between amplitude (A) and elasticity is non-linear, defined by the equation:

$$CE = \frac{100 \times A}{100 - A} \quad (3.1)$$

Consequently, the maximum clot elasticity (MCE) is calculated as:

$$MCE = \frac{100 \times MCF}{100 - MCF} \quad (3.2)$$

where MCF represents maximum clot firmness [13]. The platelet component is subsequently derived as the differential between the unperturbed and perturbed states:

$$MCE_{\text{platelet}} = MCE_{\text{EXTEM}} - MCE_{\text{FIBTEM}} \quad (3.3)$$

[13]. Because P_a is mathematically zeroed in FIBTEM, the resulting amplitude parameters demonstrate strong correlation with absolute fibrinogen concentrations, as validated against the Clauss assay with Spearman correlation coefficients above 0.80 in patient populations [14, 13].

3.3 APTEM and Hyperfibrinolysis

The APTEM assay introduces aprotinin to inhibit plasmin, effectively nullifying the plasmin-mediated fibrin degradation rate constant in the ODE system [15]. By comparing APTEM to EXTEM, the mathematical model isolates the pathological overactivation of fibrinolysis. The pathological state of hyperfibrinolysis is defined quantitatively by a clot lysis index at 30 minutes (LY_{30}) of $\geq 3\%$ [16, 17].

When stratified by the percentage of fibrinolysis at 30 minutes, the model categorizes the postinjury phenotype into three distinct mathematical presets: hyperfibrinolysis ($\geq 3\%$), physiologic ($0.081 - 2.9\%$), and shutdown ($0 - 0.08\%$) [16]. A severe manifestation of this pathology is identified as the “Death Diamond” phenotype, characterized by a viscoelastic tracing exhibiting a time to maximum amplitude (TMA) of ≤ 14 minutes and a complete return to baseline amplitude within a time to total lysis (TTL) of < 30 minutes from maximum amplitude [17]. This mathematical preset of rapid clot formation followed by immediate exponential decay of clot amplitude carries a 100% positive predictive value for mortality [17].

3.4 Dilutional Coagulopathy and Hypothermia

Pathological environmental and iatrogenic perturbations, specifically dilutional coagulopathy and hypothermia, mathematically alter the initial state variables and kinetic rate constants of the baseline ODEs [18]. Dilution with crystalloid or colloid solutions proportionally reduces the initial concentration variables for coagulation factors and fibrinogen, while simultaneously altering the rate constants governing fibrin network formation. Hypothermia, modeled as a reduction in the system temperature variable T , independently impairs the kinetic rate constants for enzymatic activity and platelet activation [18].

When modeled concurrently, dilution and hypothermia synergistically impair the clot propagation variables, specifically prolonging the clotting time (CT) and clot formation time (CFT), while reducing the alpha angle (α) [18]. The mathematical interaction demonstrates that synthetic colloid dilution interacts with hypothermia to further depress the rate constants of clot amplification beyond the additive effects predicted by isolated perturbations [18].

3.5 Antifibrinolytic Intervention

The mathematical modeling of antifibrinolytic intervention focuses on the pharmacological inhibition of the plasmin formation rate constant. Tranexamic acid (TXA), a synthetic analog of lysine, is modeled to bind reversibly to lysine receptor sites on plasminogen, functionally blocking the conversion of plasminogen to plasmin and preventing clot breakdown [19]. In the ROTEM ODE framework, the introduction of TXA sets the rate of plasmin generation, k_{plasmin} , to a negligible value.

This intervention is experimentally validated via the T-APTEM channel, where TXA is added to the whole blood sample [19]. By comparing NATEM (non-citrated whole blood without chemical activators) to T-APTEM, the model quantifies the degree to which naturally occurring fibrin degradation is suppressed. Under conditions where the maximum lysis (ML) remains within physiological limits ($< 15\%$), the mathematical difference between the NATEM and T-APTEM state trajectories is minimal, indicating the absence of a hyperfibrinolytic state requiring pharmacological suppression [19].

Numerical Implementation and Computational Architecture

4.1 Introduction to Numerical Integration in Hematology

The reduced-order ordinary differential equation (ODE) systems defined in Chapter 2, encompassing thrombin generation (Hockin et al., 2002), fibrin polymerization (Weisel & Nagaswami, 1992), fibrinolysis (Ouedraogo et al., 2024), and platelet activation (Fogelson et al., 2012), constitute a coupled, highly nonlinear dynamical system. The numerical integration of these equations presents significant computational challenges, primarily due to mathematical stiffness. As demonstrated by Danforth et al. (2009), coagulation models exhibit stiffness arising from vastly varying timescales of enzymatic reactions. The eigenvalue spectrum of the Jacobian matrix associated with the ODE system displays a high condition number, with positive and negative real parts dictating the simultaneous presence of fast and slow transient modes (Danforth et al., 2009).

This stiffness is mechanistically driven by rate constants spanning multiple orders of magnitude. For example, the rapid inhibition of thrombin by antithrombin III occurs on a timescale of seconds, whereas slow fibrin polymerization and protofibril lateral aggregation extend over minutes (Hockin et al., 2002; Weisel & Nagaswami, 1992). Explicit integration methods, such as the fourth-order Runge-Kutta method utilized in early polymerization models (Weisel & Nagaswami, 1992), require prohibitively small time steps to maintain numerical stability when applied to the full coagulation cascade. Consequently, robust implicit or semi-implicit numerical solvers are required to traverse the stiff differential equations governing hemostasis (Danforth et al., 2009).

4.2 Solver Selection and Configuration

To address the stiffness inherent in the unified hemostatic system, numerical integration is executed using the `scipy.integrate.solve_ivp` function in Python. The solver is configured to employ either the Backward Differentiation Formula (BDF) or the Livermore Solver for Ordinary Differential Equations with Automatic method switching (LSODA). LSODA adaptively alternates between stiff and non-stiff methods, optimizing computational efficiency while preserving accuracy across the distinct phases of coagulation. This approach extends the methodologies of previous studies that relied on stiff solvers such as MATLAB's `ode15s` to integrate coagulation ODEs (Danforth et al., 2009).

Strict absolute and relative tolerances are imposed to enforce mass conservation and prevent the emergence of negative concentrations, which are physically impossible but can arise during stiff integration steps. The absolute tolerance is set relative to the lowest species concentrations, such as the picomolar tissue factor concentrations used to initiate the cascade (Hockin et al., 2002), while the relative tolerance bounds the local truncation error in macroscopic species like fibrinogen.

```

1 from scipy.integrate import solve_ivp
2
3 def coagulation_ode(t, y, params):
4     # Reduced-order ODE system from Chapter 2
5     # Encompassing thrombin generation, fibrin polymerization,
6     # fibrinolysis, and platelet activation
7     dydt = compute_hemostasis_derivatives(t, y, params)
8     return dydt
9
10 def execute_simulation(y0, t_span, params):
11     solution = solve_ivp(
12         coagulation_ode,
13         t_span,
14         y0,
15         method='LSODA', # Alternately: 'BDF'
16         args=(params,),
17         rtol=1e-8,
18         atol=1e-12,
19         max_step=5e-1,
20         dense_output=True
21     )
22     return solution

```

LISTING 1. STIFF ODE INTEGRATION VIA LSODA

The parameter `max_step` is constrained to 5×10^{-1} seconds, mirroring the variable step-size limitations required to capture the rapid transitions during the propagation phase of thrombin generation without compromising the dense output resolution required for frontend visualization (Danforth et al., 2009).

4.3 Software Architecture

The computational framework mandates a strict decoupling of the scientific core from the user interface. The ODE engine, containing the deterministic mathematical model, assay perturbations (EXTEM, FIBTEM, APTEM), and pathological presets defined in Chapter 3, operates as an isolated, version-controlled backend module. This separation ensures that the mathematical computations remain entirely deterministic and independent of frontend visualization parameters, such as playback speed or graphical rendering limits.

While deterministic equivalents are preferred for real-time simulation and backend ODE integration (Ouedraogo et al., 2024), the architecture accommodates future expansions into stochastic modeling, such as the Gillespie algorithm applied to single-fibrin-fiber fibrinolysis

(Ouedraogo et al., 2024). The user interface, constructed via a React or Streamlit frontend, serves strictly as a data visualization and input transmission layer. It issues structured API requests containing assay parameters and initial conditions to the backend, which subsequently returns the integrated time-series data. By isolating the numerical core, the architecture guarantees that modifications to the visualization layer cannot perturb the underlying biochemical kinetics, an approach conceptually aligned with the separation of spatial flow models and kinetic solvers in patient-specific modeling (Ratto et al., 2021).

4.4 Reproducibility Protocol

Absolute computational reproducibility is enforced through the mandatory generation of a JavaScript Object Notation (JSON) configuration file for every simulation. This protocol ensures that any reported result, whether representing a standard EXTEM assay or a complex pathological perturbation, can be independently reconstructed. The JSON output must comprehensively document the model version, the complete parameter set, the specific assay definition, solver tolerances, and a random seed.

While the current LSODA ODE implementation is purely deterministic, the framework mandates the inclusion of a random seed in the manifest to ensure strict reproducibility for future iterations of the software that may incorporate stochastic noise models or Gillespie algorithm extensions.

```

1  import json
2
3  def generate_reproducibility_manifest(sim_id, params, assay, solver_config):
4      manifest = {
5          "simulation_id": sim_id,
6          "model_version": "v2.4.1", # Version-controlled mathematical model
7          "parameter_set": params,    # Kinetic rate constants and initial
              concentrations
8          "assay_definition": assay,  # e.g., EXTEM, FIBTEM, APTM presets
9          "solver_tolerances": {
10             "method": solver_config['method'],
11             "rtol": solver_config['rtol'],
12             "atol": solver_config['atol'],
13             "max_step": solver_config['max_step']
14         },
15         "random_seed": solver_config.get('random_seed', None) # Included
              for future stochastic extensions
16     }
17     return json.dumps(manifest, indent=4)

```

LISTING 2. REPRODUCIBILITY MANIFEST GENERATOR

This systematic documentation mitigates the uncertainties inherent in computational hematology, where variations in rate constants and solver configurations can drastically alter predicted thrombin generation profiles (Danforth et al., 2009). By binding every simulation output to a rigorous manifest, the framework adheres to the highest standards of scientific computing and

patient-specific modeling fidelity.

Model Validation and Trajectory Analysis

5.1 Introduction to Phenotypic Validation

The validation of computational models of rotational thromboelastometry (ROTEM) presents a distinct epistemological challenge relative to that of open-source biochemical simulations. Commercial ROTEM devices operate as proprietary measurement systems whose internal signal processing algorithms, transducer calibration constants, and firmware-level filtering mechanisms are not publicly disclosed. The viscoelastic amplitude $A(t)$ produced by these instruments cannot be exactly reverse-engineered from first principles, as the internal mechanical-to-electrical transduction pathway, including the specific torsion constant of the elastic coupling wire, the analog-to-digital conversion parameters, and the proprietary smoothing kernels, remains opaque to independent verification.

Given this constraint, model validation proceeds through phenotypic agreement rather than exact quantitative matching. The simulated trajectory $A(t)$, derived from the reduced-order ordinary differential equation (ODE) system specified in Chapter 2 and integrated via the Livermore Solver for Ordinary Differential Equations with Automatic method switching (LSODA) described in Chapter 4, is assessed for its capacity to reproduce recognizable viscoelastic curve morphologies and parameter ranges documented in the clinical literature. This approach treats the ROTEM device as a transfer function whose output morphology, rather than its absolute internal state, constitutes the quantity of interest. Validation thus consists of demonstrating that perturbations to the model's state variables and rate constants produce directionally correct and clinically consistent changes in the standard ROTEM reporting parameters: clotting time (CT), maximum clot firmness (MCF), clot lysis index at 30 minutes (CLI30), and the derived clot elasticity (CE) metrics.

5.2 Normal EXTEM and FIBTEM Baseline

Calibration of the reduced-order model to physiological baseline conditions requires that the simulated amplitude trajectory $A(t)$ satisfy established reference ranges for both the EXTEM and FIBTEM assays under normal initial conditions. The EXTEM assay, initiated by exogenous tissue factor, reflects the integrated contributions of thrombin generation, fibrinogen polymerization, and platelet activation to clot firmness. The FIBTEM assay, in which cytochalasin D abolishes the platelet contribution by setting the platelet activation state variable P_a to zero (as defined in Chapter 3, Section 3.2), isolates the fibrinogen-dependent component of clot strength.

Reference ranges for ROTEM parameters have been established through multi-centre in-

vestigation. As reported by Lang et al. [12], normal baseline clot firmness parameters were established for whole blood, platelet-rich plasma, and platelet-poor plasma, with cytochalasin D completely abolishing the platelet contribution to clot strength. De Vries et al. [14] reported median FIBTEM MCF values of 15.0 mm [interquartile range, IQR: 12.0–18.0 mm] in a cohort of 75 healthy subjects, with manufacturer-specified normal ranges of 9–25 mm for MCF and 7–23 mm for A10. Levrat et al. [15] reported EXTEM MCF reference ranges of 52–66 mm and FIBTEM MCF reference ranges of 8–20 mm, derived from healthy volunteer controls. These quantitative reference ranges serve as calibration targets for the model’s baseline parameter set.

The model is calibrated such that, under baseline initial conditions, the simulated EXTEM trajectory $A(t)$ attains an MCF of approximately 60 mm, with a CT of approximately 50 seconds, values consistent with the reference ranges reported by Levrat et al. [15] (EXTEM CT: 30–81 s; EXTEM MCF: 52–66 mm). Simultaneously, the simulated FIBTEM trajectory, generated by setting $P_a = 0$, attains an MCF of approximately 15 mm, consistent with the healthy-subject median reported by de Vries et al. [14].

The mechanical-to-signal mapping function defined in Chapter 2, Section 2.3:

$$A(t) = A_{\max} \cdot \frac{[F_p](t)}{[F_p](t) + K_F} \cdot \frac{[P_a](t)}{[P_a](t) + K_{P_a}} \quad (5.1)$$

is calibrated such that the half-saturation constants K_F and K_{P_a} , along with $A_{\max}$, yield these target amplitudes under normal physiological initial conditions for fibrinogen concentration (median 2.7 g/L; de Vries et al. [14]) and platelet count. The multiplicative form ensures that the FIBTEM trajectory, with $[P_a] = 0$, produces negligible amplitude unless the fibrin polymer term alone is sufficient to generate measurable clot firmness, a behavior consistent with the observation that FIBTEM clot firmness correlates with fibrinogen concentration rather than platelet count [14, 13].

5.3 Validation of Pathological Presets

5.3.1 Hypofibrinogenemia

Hypofibrinogenemia, defined as a plasma fibrinogen concentration below 1.5 g/L, represents a state in which the structural substrate for clot formation is depleted. De Vries et al. [14] demonstrated a strong linear correlation between FIBTEM clot firmness parameters (MCF and A10) and plasma fibrinogen concentration measured by the Clauss assay, with Spearman correlation coefficients (r_s) exceeding 0.80 across all patient subgroups and $r_s = 0.66$ ($p < 0.001$) in healthy subjects. The study identified optimal FIBTEM MCF cut-off values of ≤ 10.5 mm for predicting fibrinogen levels below 1.5 g/L, with a sensitivity of 88.9% and specificity of 88.1%. The finding that reducing fibrinogen directly reduces FIBTEM amplitude provides a direct mechanistic correlate for model perturbation.

In the reduced-order model, hypofibrinogenemia is simulated by proportionally reducing the initial fibrinogen state variable $[Fg]_0$. Because the FIBTEM assay isolates the fibrinogen component by setting $P_a = 0$, the resulting $A(t)$ trajectory depends exclusively on the fibrin polymer concentration $[F_p](t)$, which is in turn determined by the thrombin-catalyzed conversion of $[Fg]$ as governed by the ODE $d[Fg]/dt = -\alpha_1 \cdot [IIa] \cdot [Fg]$ (Chapter 2, Section

2.2.2). Reducing $[Fg]_0$ proportionally reduces the maximum achievable $[F_p]$, and through the saturable mapping function, proportionally reduces the FIBTEM MCF. This behavior is consistent with the linear correlation structure reported by de Vries et al. [14], who observed that reducing fibrinogen directly reduces FIBTEM amplitude.

The model predicts that a reduction of $[Fg]_0$ to approximately 55% of baseline (corresponding to a fibrinogen level of approximately 1.5 g/L from a baseline of approximately 2.7 g/L) yields a FIBTEM MCF of approximately 10–11 mm, consistent with the empirically determined cut-off of ≤ 10.5 mm [14]. The EXTEM trajectory under the same perturbation exhibits a less pronounced reduction in MCF, because the platelet contribution to clot elasticity, mediated by $[P_a]$, remains intact. This differential response between EXTEM and FIBTEM is mechanistically consistent with the clinical observation that FIBTEM parameters are more sensitive to fibrinogen depletion than EXTEM parameters, which retain a platelet-dependent component [13].

5.3.2 Thrombocytopenia

Thrombocytopenia, characterized by a reduction in circulating platelet count, is simulated in the model by reducing the initial platelet state variable $[PL_{\text{total}}]_0$, which proportionally reduces the maximum achievable activated platelet concentration $[P_a]$. Solomon et al. [13] established that the platelet component of clot strength is calculated as the difference in maximum clot elasticity between the EXTEM and FIBTEM assays:

$$\text{MCE}_{\text{platelet}} = \text{MCE}_{\text{EXTEM}} - \text{MCE}_{\text{FIBTEM}} \quad (5.2)$$

where $\text{MCE} = (100 \times \text{MCF}) / (100 - \text{MCF})$.

Solomon et al. [13] emphasized that this calculation must be performed using clot elasticity rather than clot amplitude, owing to the nonlinear relationship between amplitude and elasticity. A specific increment of amplitude (ΔA) yields different increments of elasticity (ΔCE) depending on the initial amplitude value. The theoretical data presented by Solomon et al. [13] demonstrate that progressive reduction in platelet count (from 100,000/ μL to 10,000/ μL) reduces EXTEM MCE while FIBTEM MCE remains constant, as the latter is determined solely by fibrinogen. At a platelet count of 10,000/ μL , the EXTEM MCF was reduced to 30 mm (MCE = 43), while the FIBTEM MCF remained at 0 mm (MCE = 0), yielding a platelet component of 43.

In the reduced-order model, reducing $[PL_{\text{total}}]_0$ directly attenuates the $[P_a]$ term in the mechanical-to-signal mapping function. Because the FIBTEM assay sets $[P_a] = 0$ regardless of platelet count, the FIBTEM trajectory remains unaffected by thrombocytopenia. The EXTEM trajectory, however, exhibits a reduction in MCF proportional to the reduction in $[P_a]$, isolating the platelet contribution to clot strength. This behavior is consistent with the clinical observation that thrombocytopenia reduces EXTEM clot strength while FIBTEM remains primarily driven by fibrinogen [13].

The model's capacity to independently modulate the platelet state variable while preserving the fibrinogen trajectory provides a mechanistic basis for the differential diagnosis of thrombocytopenia versus hypofibrinogenemia using paired EXTEM/FIBTEM analysis. This differential diagnostic capability is a direct consequence of the multiplicative structure of the mapping function, which requires both fibrin polymer and activated platelets for full clot strength; the absence of either component yields attenuated amplitude.

5.3.3 Hyperfibrinolysis and the “Death Diamond” Phenotype

Hyperfibrinolysis (HF), defined as a clot lysis index at 30 minutes (LY30) of $\geq 3\%$ [17], represents a state of pathological overactivation of the fibrinolytic cascade. In the reduced-order model, HF is simulated by elevating the initial tissue plasminogen activator (tPA) and plasmin state variables, which increases the rate of plasmin-mediated fibrin degradation as governed by the deterministic equivalent ODE:

$$\frac{d[F_{\text{intact}}]}{dt} = -k_{\text{deg}} \cdot [\text{Pln}] \cdot ([N_{00}] + [N_{03}]) \quad (5.3)$$

defined in Chapter 2, Section 2.2.3, where k_{deg} is the plasmin-mediated rate of fibrin degradation, $[\text{Pln}]$ is the plasmin concentration, and the term $([N_{00}] + [N_{03}])$ represents the concentration of exposed, undegraded binding sites accessible to plasmin. The fiber is considered cleaved when two-thirds of the binding doublets within the cross-section have been degraded.

Chapman et al. [17] identified a specific hyperfibrinolytic phenotype termed the “Death Diamond,” characterized by a viscoelastic tracing exhibiting a time to maximum amplitude (TMA) of ≤ 14 minutes and a complete return to baseline amplitude within a time to total lysis (TTL) of < 30 minutes from maximum amplitude. This pattern, observed in 14 of 572 trauma patients, carried a 100% positive predictive value for mortality. The trajectory morphology consists of a steep ascent to maximum amplitude followed by rapid exponential decay to baseline, producing a diamond-shaped curve.

When the model’s tPA/plasmin variables are elevated, the simulated $A(t)$ trajectory reproduces this characteristic morphology: an initial rapid ascent driven by the propagation phase of thrombin generation and fibrin polymerization, followed by an abrupt descent as plasmin-mediated degradation of intact fibrin outpaces ongoing polymerization. The rate of amplitude decay is governed by k_{deg} and the plasmin concentration $[\text{Pln}]$, with the fiber cleavage threshold determining the temporal offset between plasmin exposure and measurable amplitude loss. Baseline parameter values of $k_{\text{deg}} = 5 \text{ s}^{-1}$ and $k_{\text{exp}} = 5 \text{ s}^{-1}$, as specified in Chapter 2 from Ouedraogo et al. [7], constrain the degradation kinetics to physiologically plausible timescales.

The APTEM assay, which introduces aprotinin to inhibit plasmin (as defined in Chapter 3, Section 3.3), is modeled by nullifying the plasmin-mediated degradation rate constant in the ODE system. Levrat et al. [15] demonstrated that APTEM reverses the hyperfibrinolytic phenotype observed in EXTEM, confirming that the lysis is plasmin-mediated. In the model, setting the degradation rate to zero in the APTEM configuration prevents the decay phase of $A(t)$, causing the trajectory to sustain its maximum amplitude, a behavior consistent with the clinical observation that aprotinin restores clot firmness in hyperfibrinolytic samples [15].

Levrat et al. [15] reported that in patients with hyperfibrinolysis, the EXTEM MCF was reduced to a median of 11 mm [IQR: 10–13 mm], while the APTEM MCF was preserved at 23 mm [IQR: 17–23 mm], representing a ΔMCF of $> 7\%$ (calculated as $[(\text{MCF}_{\text{APTEM}} - \text{MCF}_{\text{EXTEM}}) / \text{MCF}_{\text{EXTEM}}] \times 100$). The diagnostic threshold of $\Delta\text{MCF} > 7\%$ yielded a sensitivity of 80% and specificity of 100% for hyperfibrinolysis. The model reproduces this differential: the EXTEM trajectory exhibits premature amplitude decay due to ongoing plasmin activity, while the APTEM trajectory sustains clot firmness. The magnitude of the EXTEM–APTEM differential scales with the plasmin concentration, providing a quantitative basis for the diagnostic threshold established by Levrat et al. [15].

5.4 Error Metrics and Limitations

Quantitative assessment of the deviation between simulated and clinically observed trajectories requires metrics that accommodate both point-wise amplitude differences and temporal misalignment. Two complementary metrics are employed: Root Mean Square Error (RMSE) and Dynamic Time Warping (DTW).

RMSE provides a point-wise measure of amplitude deviation:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (A_{\text{sim}}(t_i) - A_{\text{clin}}(t_i))^2} \quad (5.4)$$

where $A_{\text{sim}}(t_i)$ and $A_{\text{clin}}(t_i)$ are the simulated and digitized clinical amplitudes at time point t_i , and N is the number of sampled time points. RMSE penalizes deviations in both amplitude and temporal phase equally, rendering it sensitive to misalignment in the clotting time (CT) parameter. Clinical curves for comparison are obtained by digitizing published ROTEM tracings from the clinical literature [12, 15, 17] using semi-automated extraction tools.

DTW addresses the limitation of RMSE by computing an optimal non-linear alignment between the simulated and clinical time series, permitting comparison of trajectories that exhibit similar morphology but are temporally offset. The DTW distance is computed as the minimal cumulative cost of aligning the two sequences under a set of monotonicity constraints. This metric is particularly relevant for ROTEM validation, where inter-patient variability in CT (ranging from 30–81 seconds in normal subjects; Levrat et al. [15]) can produce significant temporal shifts in trajectory onset without altering the characteristic curve morphology. By permitting elastic temporal alignment, DTW isolates morphological deviations from temporal phase shifts, enabling independent assessment of clot formation kinetics and clot structural integrity.

Several limitations of the reduced-order modeling approach must be acknowledged. First, the model assumes spatial homogeneity of the coagulation process within the ROTEM cuvette, neglecting spatial gradients in reactant concentration that may arise from diffusion-limited transport of large macromolecular species (e.g., fibrinogen, plasminogen) and from the heterogeneous distribution of platelets within the clotting matrix. Commercial ROTEM devices impose a low-shear oscillatory regime that minimizes convective mixing; under these conditions, spatial heterogeneity in clot structure may contribute to deviations between simulated and observed trajectories that cannot be captured by the lumped-parameter ODE framework. As noted in Chapter 4, Section 4.3, the computational architecture is conceptually aligned with the separation of spatial flow models and kinetic solvers in patient-specific modeling [8], but the current implementation does not incorporate spatial resolution.

Second, the model's mechanical-to-signal mapping function assumes a multiplicative interaction between fibrin polymer and activated platelets, which represents a phenomenological simplification of the complex biomechanical interactions, including platelet-mediated clot retraction, fibrin fiber cross-linking by factor XIII, and the viscoelastic contributions of red blood cells, that collectively determine the mechanical properties of the formed clot. Solomon et al. [13] noted that although red blood cells comprise over 90% of blood clot volume, clot strength is derived from the interaction of the fibrin network and platelets, and that the relationship between clot amplitude and clot elasticity is nonlinear. The model's use of the CE transformation,

$CE = (100 \times A)/(100 - A)$, partially addresses this nonlinearity in the reporting domain, but the underlying assumption of a static, concentration-dependent mapping from biochemical state to mechanical output neglects the time-dependent viscoelastic creep and stress relaxation behaviors of the clot matrix.

Third, the model does not explicitly represent the effects of hematocrit on viscoelastic measurements. Red blood cells contribute to the physical properties of the clot through mechanisms distinct from those of fibrin and platelets [13], and their influence on $A(t)$ is not captured by the current state variables.

Fourth, the parameter set used for calibration is derived from heterogeneous experimental conditions across multiple laboratories, with kinetic rate constants exhibiting variability of up to 1,000-fold across investigations (as noted in Chapter 2, Section 2.4). While the sensitivity analysis framework established by Danforth et al. [2] identifies a small subset of parameters, particularly those governing the initiating TF·VIIa complex and thrombin inhibition, as disproportionately influential, the residual uncertainty in less-sensitive parameters may propagate through the nonlinear ODE system to produce unquantified effects on trajectory morphology. Danforth et al. [2] demonstrated that uncertainty in the rate constant governing thrombin inhibition by antithrombin III produced effects equivalent to the combined contribution of the 19 least influential rate constants, indicating that even isolated parameter uncertainty can substantially alter predicted thrombin generation profiles.

Despite these limitations, the phenotypic validation approach, grounded in qualitative and semi-quantitative agreement with published clinical curves, provides a defensible basis for the model's utility as a computational framework for ROTEM trajectory analysis. The capacity of the model to reproduce directionally correct responses to defined perturbations of fibrinogen, platelet, and plasmin state variables supports its use as an *in silico* platform for hypothesis generation and assay optimization. The model correctly predicts that fibrinogen depletion preferentially reduces FIBTEM MCF [14], that platelet depletion preferentially reduces EXTEM MCF while sparing FIBTEM [13], and that elevated plasmin activity produces the characteristic “Death Diamond” trajectory reversible by aprotinin [17, 15]. Absolute quantitative prediction of patient-specific trajectories, however, requires additional parameter identification from individual patient data and integration of patient-specific initial conditions into the ODE framework.

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